

Project Proposal

BinSense

Smart Waste Management System using IoT and AI

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Supervisor Meeting Log:

No.	Discussion	Task Assignmnet	Supervisor Signature
Meeting 01 Date: 20 - 03 – 2026	New FYP Ideas, Feasibility, Constraints	Write Initial Proposal	
Meeting 02 Date: 2 - 04 – 2026	Modules to include, Stack selection.	Finalize The Modules and scope document	

1. Introduction

Rapid urbanization in Pakistan has led to a significant increase in waste generation, creating challenges in efficient waste collection and management. Traditional waste collection systems rely on fixed schedules, which often result in overflowing bins or unnecessary collection of half-empty bins. This leads to inefficient resource utilization, increased operational costs, and environmental pollution.

To address these issues, this project proposes a **Smart Waste Management System using IoT and AI**, which monitors bin levels in real-time, optimizes collection routes, and predicts future waste levels. The system integrates sensors, cloud computing, and intelligent algorithms to improve efficiency and sustainability.

2. Related System Analysis / Literature Review

Existing System	Weakness	Proposed Solution
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Traditional Waste Collection	Fixed schedule, no real-time data	Real-time bin monitoring
Basic Smart Bins	Only show fill level	Adds routing + prediction
GPS Tracking Systems	No optimization	Route optimization added
Manual Monitoring	Human-dependent	Fully automated IoT system

3. Problem Statement

Urban waste management systems today face a “triple challenge” of inefficiency, lack of monitoring, and poor planning:

- **No Real-Time Monitoring:** Authorities do not know which bins are full and which are empty. This leads to overflowing garbage bins, causing bad odor, pollution, and health risks for the public.
- **Inefficient Collection Routes:** Waste collection vehicles follow fixed routes instead of optimized ones. This results in unnecessary fuel consumption, increased costs, and wasted time.
- **Lack of Data & Planning:** There is no proper use of historical data to understand waste patterns. Without prediction, the system reacts only after problems occur instead of preventing them.
- **Manual System Dependency:** The current system relies heavily on human observation, which is slow, error-prone, and not scalable for growing cities.

4. Problem Solution / Objectives of the Proposed System

The solution to these problems is the development of a “**Smart Waste Management System.**” This system will use IoT sensors to monitor bin levels in real-time and send data to a centralized platform.

The system will also include intelligent route optimization to ensure that only necessary bins are collected, reducing fuel usage and operational costs. In addition, an AI-based prediction model will analyze historical data to forecast when bins are likely to become full, enabling proactive waste management.

Overall, the system will transform waste collection from a manual and reactive process into an automated and data-driven smart solution.

4.1 Objectives

- **Real-Time Monitoring:** To track bin fill levels using IoT sensors and reduce overflow situations.
- **Cost Optimization:** To minimize fuel consumption and operational costs through route optimization.
- **Efficient Collection:** To ensure workers collect only necessary bins instead of following fixed routes.
- **Data-Driven Decisions:** To store and analyze historical data for better planning and insights.
- **Predictive Management:** To forecast future bin status using AI and improve waste collection scheduling.
- **System Accessibility:** To provide an easy-to-use dashboard for admins and workers.

5. Scope of the Project

The scope of this project focuses on building a complete smart waste management solution for urban areas.

- **What we include:** IoT-based bin monitoring, real-time dashboard, route optimization, data storage and analytics, and AI-based prediction of bin fill levels.
- **What we don't include:** We will not manufacture physical bins or large-scale hardware infrastructure. The system will focus mainly on software integration and sensor-based monitoring.
- **User Base:** The system is designed for municipal authorities, waste collection workers, and city management departments.

6. Modules

6.1 User Management Module

- User registration and login
- Role-based access:
 - Admin
 - Worker
- Profile management

6.2 Smart Bin Monitoring Module

- Uses ultrasonic sensor to measure distance
- Converts distance → fill percentage
- Sends data via WiFi/GSM to server

6.3 Data Processing & Storage Module

- Receives sensor data through API
- Stores data in database
- Updates system in real-time

6.4 Bin Status Classification Module

Classifies bins into:

- Empty (0–30%)
- Medium (30–70%)
- Full (70–100%)

6.5 Alert & Notification Module

- Sends alerts when bins are full
- Notifies:
 - Admin
 - Workers

6.6 Route Optimization Module

- Selects only FULL bins
- Calculates shortest path using:
 - Dijkstra Algorithm OR
 - Google Maps API

6.7 Dynamic Re-routing Module:

- Adjusts routes in real-time if conditions change
- Ensure vehicles adapt if new information arrives.

6.8 Task & Collection Management Module

- Assigns tasks to workers
- Updates collection status
- Tracks completed jobs

6.9 Dashboard & Visualization Module

- Map-based bin display
- Color indicators:
 - Green = Empty
 - Yellow = Medium
 - Red = Full
- Real-time updates

6.10 Analytics & Reporting Module

- Historical data analysis
- Graphs of bin usage
- Performance insights

6.11 Location Tracking Module

- GPS-based bin locations
- Display bins on map

6.12 AI-Based Prediction Module

- Predicts when bin will be full
- Suggests optimal collection time

6.13 Environmental Monitoring Module:

- Sensors (e.g. air quality or odor) to detect unhealthy conditions

- **Problem Solved:** Not critical for collection routing, but could flag hazardous bins or areas.

6.14 Public Reporting Feature

- Each bin includes a **QR code / barcode**
- Citizens can scan it using a mobile phone
- Allows public to report issues

7. System Limitations / Constraints

- **Sensor Accuracy:** The system depends on ultrasonic sensors to measure bin levels. Factors like irregular waste shapes, moisture, or sensor blockage can sometimes give slightly inaccurate readings.
- **Internet Connectivity:** The system requires WiFi or GSM connectivity to send data. In areas with poor or no network coverage, real-time updates may be delayed.
- **Hardware Dependency:** Since the system relies on physical IoT devices, any hardware failure (sensor damage, power issues) can affect data collection.
- **GPS Accuracy:** Location tracking may not always be perfectly accurate, especially in dense urban areas with weak GPS signals.
- **Prediction Limitations:** The AI model predictions depend on historical data. In the early stages, predictions may not be very accurate due to limited data availability.

8. Data Gathering Approach

Collecting accurate and useful data is essential for making the Smart Waste Management System effective. Since real-world waste data is not easily available, we followed a practical approach:

- **Sensor Data (Real-Time Data):** The main source of data comes from IoT-enabled smart bins. Ultrasonic sensors measure the fill level of bins and continuously send data to the system. This helps in real-time monitoring and decision-making.

- **Simulated & Historical Data:** In the initial phase, we may use simulated datasets to train and test the system, especially for the AI prediction module. Over time, real data collected from sensors will replace simulated data for better accuracy.
- **Location Data:** GPS modules are used to collect the exact location of bins. This data is essential for mapping and route optimization.
- **System Usage Data:** The system will also collect data on collection times, worker activity, and bin usage patterns. This helps in generating analytics and improving system performance.
- **Supervisor Guidance & Research:** We also rely on research papers and supervisor feedback to ensure that the collected data and system design are practical and feasible.

9. Tools and Technologies

Component	Technology
Frontend	React JS
Backend	Node.js / Spring Boot
Database	PostgreSQL
IoT	Arduino / ESP8266
AI Model	Python (Scikit-learn)
APIs	REST API
Maps	Google Maps API

10. Project Stakeholders

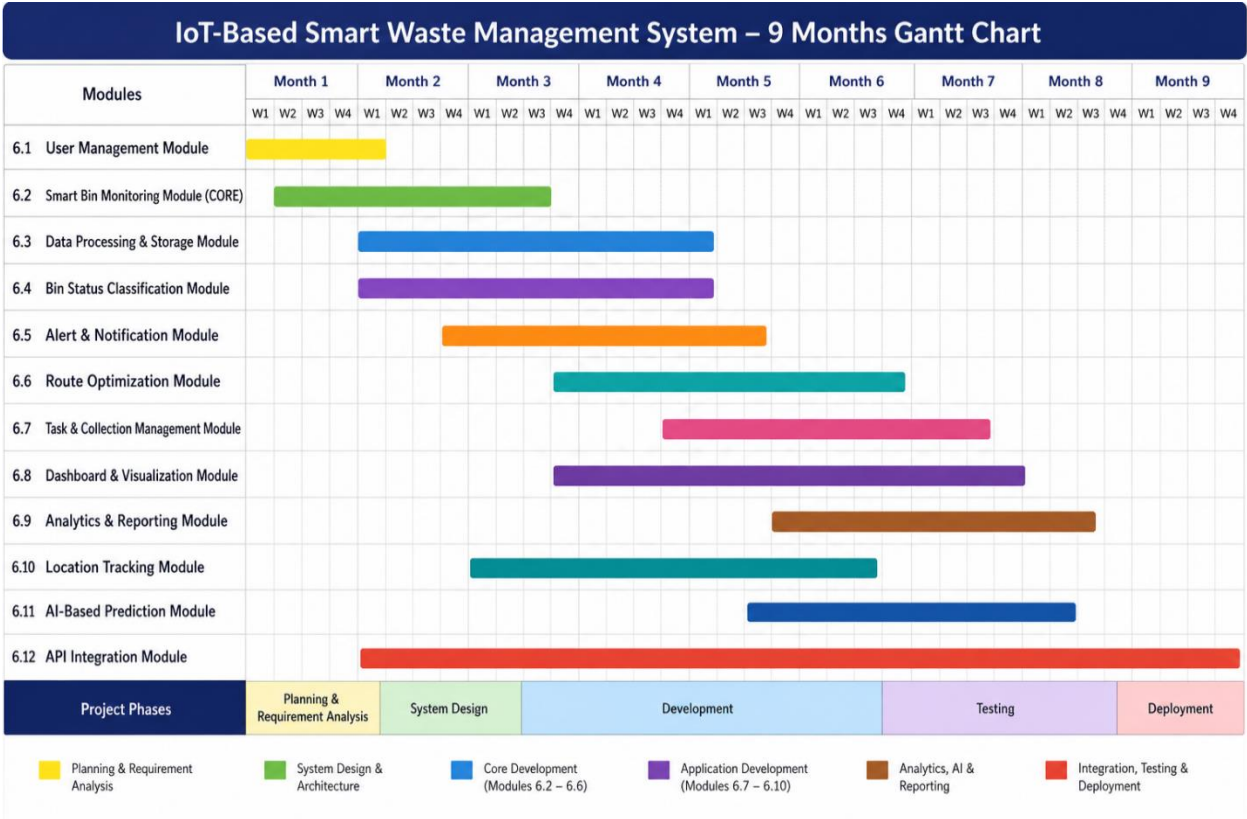
Project Sponsor	Air University Islamabad
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Stakeholder	• Muhammad Areeb Islam • Muhammad Kamran • Abdul Basit • Sir Mohabbat Ali(Supervisor) • Saaf Seher (A recycle workspace community) • Final Year Project Committee: Evaluation of project
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11. Work Division

Member	Responsibility
Muhammad Areeb Islam	Backend + API + Database
Abdul Basit	Frontend + Dashboard
Muhammad Kamran	IoT + Sensor Integration

12. Gantt Chart



13. Mockups

scope document of BinSense

User Login

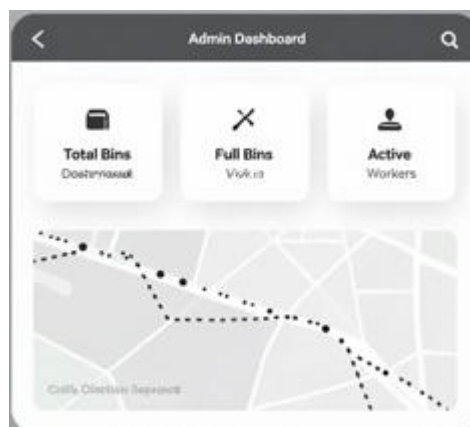


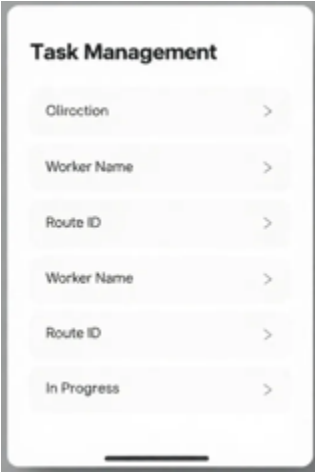
Username

Password

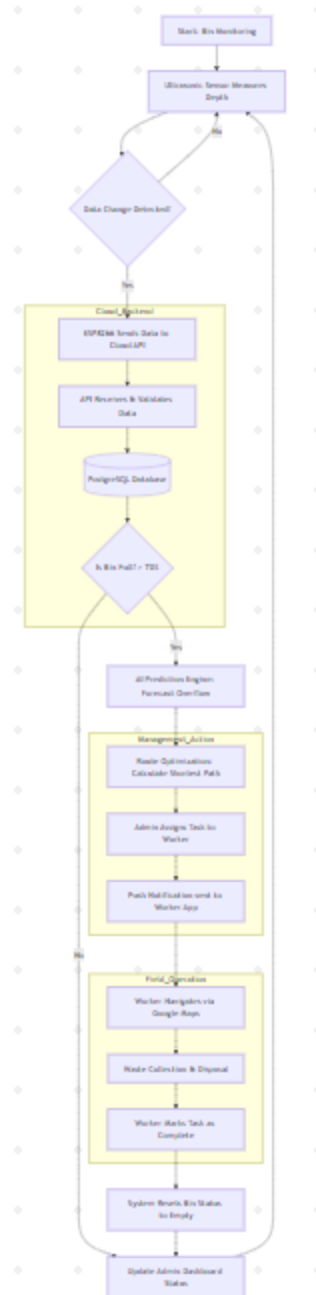
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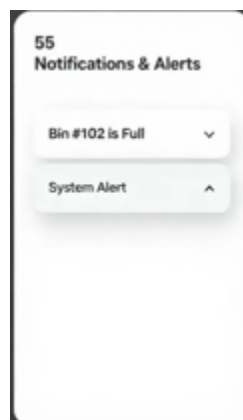
Login





scope document of BinSense





14. References

S.No	Reference	Link
1	Scikit-learn Documentation	https://scikit-learn.org/stable/
2	IoT-Enabled Smart Waste Management Systems for Smart Cities (IEEE Access)	https://doi.org/10.1109/ACCESS.2022.3188308
3	IoT-Enabled Waste Management in Smart Cities (IJACSA)	https://thesai.org/Publications/ViewPaper?Code=IJACSA&Issue=4&SerialNo=15&Volume=16
4	Sustainable Smart IoT-based Solid Waste Management System (ScienceDirect)	https://www.sciencedirect.com/science/article/pii/S0167739X24001183
5	Waste Management 2.0 using IoT for Smart Cities (PLOS ONE)	https://doi.org/10.1371/journal.pone.0307608

6	IoT Based Smart Waste Management System (arXiv)	https://arxiv.org/abs/2411.09710
7	Smart Waste Collection System – Code for Pakistan Project	https://codeforpakistan.org/projects/smart-waste-bins
8	Smart Waste Management in Canada Bay (LoRaWAN IoT Case Study)	https://www.meshed.network/lorawan-iot-smart-waste-city-of-canada-bay/
9	Saaf Sheher Smart City Waste Initiative	https://saafsheher.com/home/
10	IoT Smart City Waste Management Book (Google Books)	https://books.google.com.pk/books?hl=en&lr=&id=kmqQDwAAQBAJ